



Predicting Global Deforestation

STOR 320 FINAL PROJECT

Maanya, Kevin, Dong Cai, Siyu, Katherine

An aerial photograph showing a dense green forest in the upper right and a cleared, brownish area with many thin, vertical tree trunks in the lower left. A dark blue rectangular box with a red border is centered over the image, containing white text.

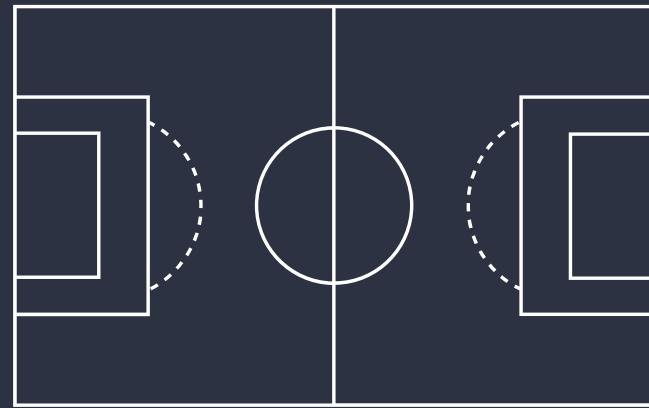
10 MILLION

Hectares of Forests are cleared every year





27



Soccer Fields

Deforested every minute

Deforestation stands as one of the most severe environmental crises facing our planet today.

DEFORESTATION **THREATENS**



2 Billion

Individuals that rely on forests for their livelihoods



250 Million

Direct forest inhabitants



> 80%

Of Earth's wildlife and biodiversity

DEFORESTATION **IMPACTS**

1

Increased GHG Emissions

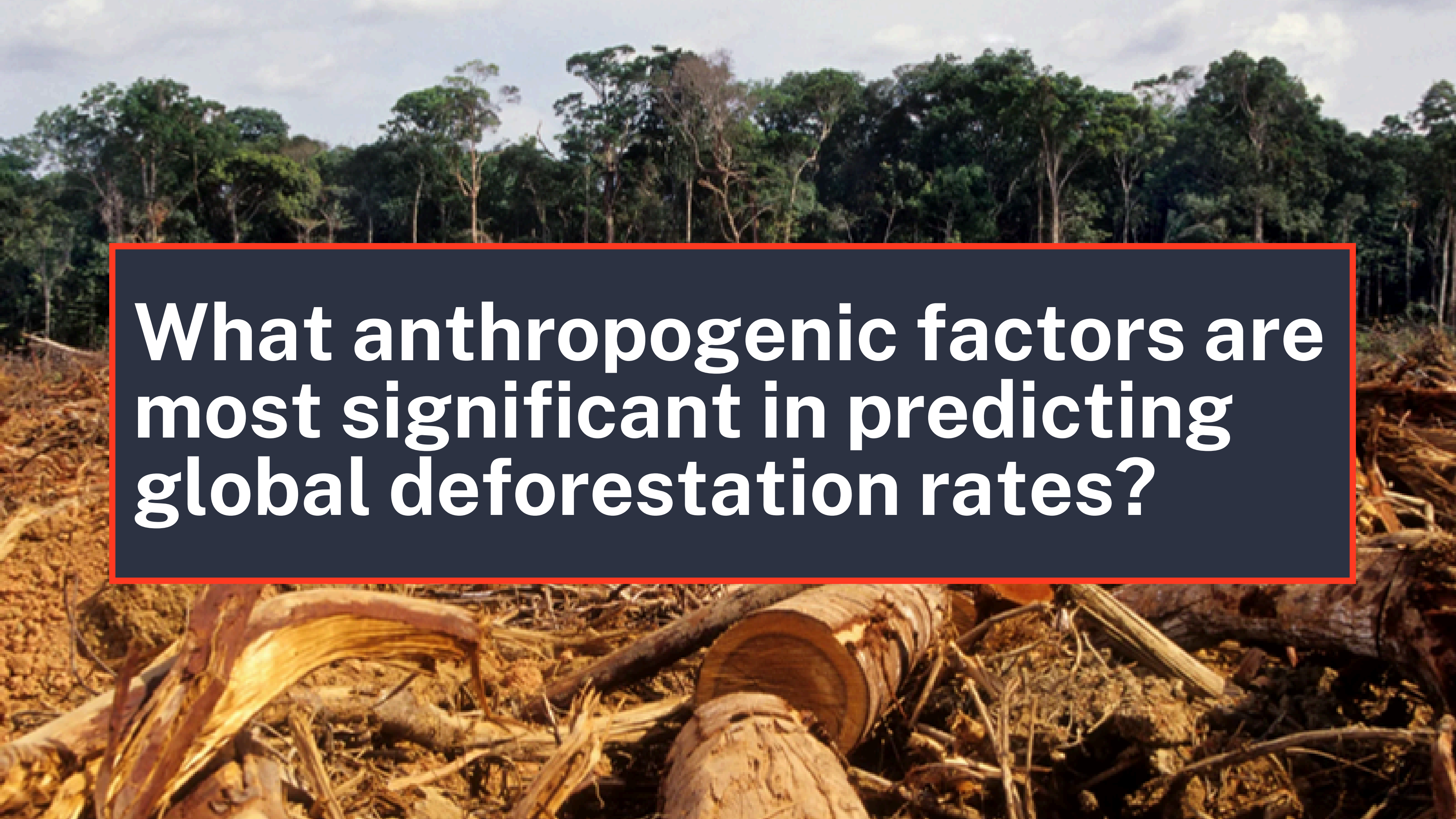
2

Decreased Biodiveristy

3

Disrupted Water Cycles

ANTHROPOGENIC

The image shows a landscape where a forest has been cleared. In the foreground, there is a large pile of cut logs and debris, including a prominent log with a visible cross-section. The ground is covered in brown, dry leaves and twigs. In the background, a dense line of green trees stands against a cloudy sky. A dark blue rectangular box with a red border is superimposed over the middle of the image, containing white text.

What anthropogenic factors are most significant in predicting global deforestation rates?

Our final data set focuses on global deforestation rates from 2001 – 2016 and various anthropogenic variables tied to deforestation.

Country	Year	Carbon emissions	Deforestation	Agriculture expansion	Meat production	Forestry production (ton)	Forestry production (volume)	Urban population (total)	Urban population (percentage)	GDP growth
Alghanistan	2001	24035.52	91.68592	216834	278743.3	83747	3474150	19088.63	22.16940	NA
Alghanistan	2002	41638.92	189.70335	238246	304321.2	86790	3510626	21000.26	22.26128	NA
Alghanistan	2003	49366.33	253.28507	219998	298619.0	89943	3548176	22645.13	22.35344	8.7
Alghanistan	2004	34118.89	207.39641	228874	322976.9	93211	3586832	23553.55	22.50042	0.7
Alghanistan	2005	42484.12	246.19251	210597	318614.7	96598	3626629	24411.19	22.70305	11.8
Alghanistan	2006	24077.37	159.04678	223714	272400.0	99572	3658194	25442.94	22.90725	5.4

KEY VARIABLES

- Tree Loss
- Carbon emissions
- Agriculture expansion
- Meat production
- Forestry production (volume) & Forestry production (sum)
- Urban population (total) & Urban population (percentage)
- GDP growth

DATA SOURCES

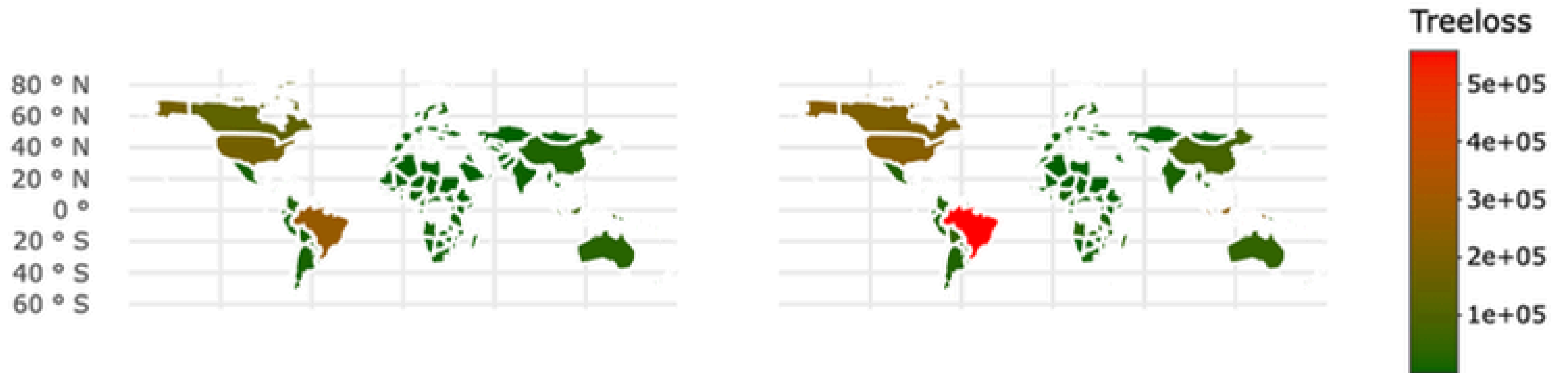
- Global Forest Watch
 - Developed by the World Resources Institute, Google, USAID
 - Open-source web application for real-time monitoring
- UN Food and Agriculture Organization
- World Bank
- Worldometer

Deforestation has increased from 2001 – 2016 in select world regions.

Deforestation Across The World

YEAR 2001

YEAR 2016



Our initial full model included all of the variables in our data set

Full Model = Deforestation ~ carbon emissions + agriculture expansion + meat production + forestry production in volume + forestry production in ton + urban population + urban population percentage + GDP growth

Select significant variables from the full model

- P-value between each predictor and **deforestation**
- Select statistically significant variables based on p-values (< 0.05)
- Remove Agricultural Expansion & GDP Growth
- **+ 6 single models with each variable**

Cleaned Model by P-value = Deforestation ~ carbon emissions + meat production + forestry production in volume + forestry production in ton + urban population + urban population percentage

Perform 10-Fold Cross Validation to see the accuracy of model candidates

- Explore the out-of-sample error using
 - **RMSE (Root Mean Squared Error)** and **MAE (Mean Absolute Error)**
- Measure the average magnitude of error
 - **RMSE** weights high errors due to outliers
 - **MAE** weights errors relatively equal

Use Backward Elimination to validate findings

- **Model selection and avoid overfitting**
- **AIC (Akaike Information Criterion)**
- **Formula: number of variables + functionality**
- **Validates the previously cleaned model finding**

Cleaned Model by P-value = Deforestation ~ carbon emissions + meat production + forestry production in volume + forestry production in ton + urban population + urban population percentage

Add interaction terms to improve the accuracy of the cleaned model

- The carbon emission model performs relatively well
- Potential interrelated forces between human activities and production factors
- Create the below three interaction models

Interaction Model 1 = Deforestation ~ carbon emissions + meat production + forestry production in volume + forestry production in ton + urban population + urban population percentage
+I(carbon emissions x meat production)

Add interaction terms to improve the accuracy of the cleaned model

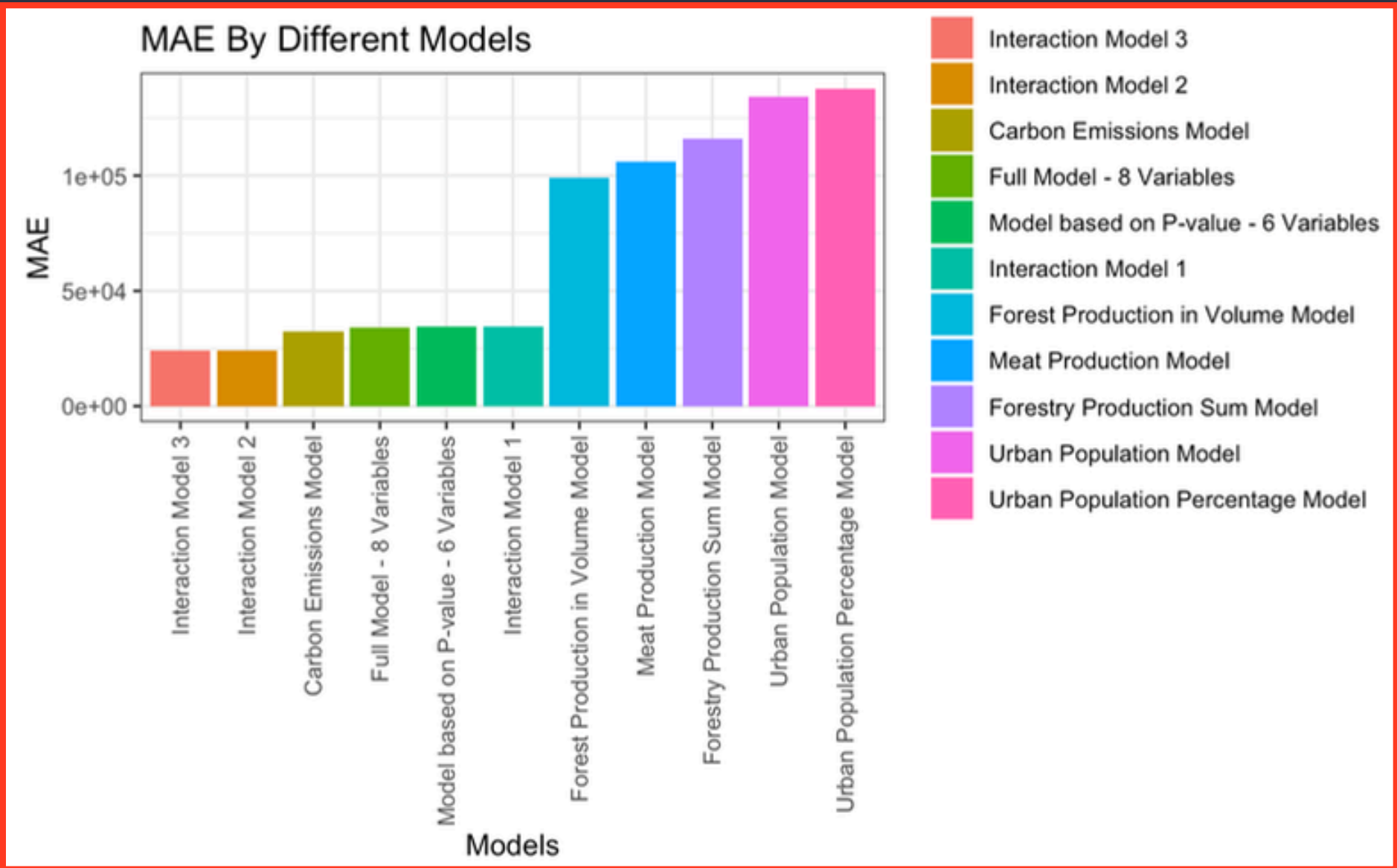
Interaction Model 2 = Deforestation ~ carbon emissions + meat production + forestry production in volume + forestry production in ton + urban population + urban population percentage + I(carbon emissions x meat production) + I(carbon emissions x urban population)

Add interaction terms to improve the accuracy of the cleaned model

Interaction Model 3 = Deforestation ~ carbon emissions + meat production + forestry production in volume + forestry production in ton + urban population + urban population percentage + **I(carbon emissions x meat production)** + **I(carbon emissions x urban population)** + **I(carbon emissions x forestry production)**

Select the model that has the lowest MAE & RMSE and complexity

MAE Values



RMSE Values



Interaction Model 2 has the best results

- Deforestation is a complex issue globally
- Interrelated forces between anthropogenic predictors of deforestation

Interaction Model 2 = Deforestation ~ carbon emissions + meat production + forestry production in volume + forestry production in ton + urban population + urban population percentage + I(carbon emissions x meat production) + I(carbon emissions x urban population)

Leverage Data to Understand and Address Human-Driven Deforestation

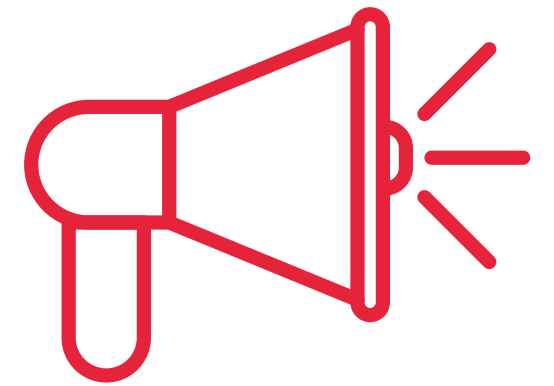
OUR **MODEL** WILL



**Inform key human
drivers of deforestation
through modeling**



**Easy, accessible
information for the
general public**



**Prompt future policy
and advocacy
initiatvies**

WORKS CITED

<https://news.un.org/en/story/2020/07/1068761#:~:text=The%20annual%20rate%20of%20deforestation,great%20concern%2C%20according%20to%20FAO.>

<https://www.worldwildlife.org/stories/what-is-forest-degradation-and-why-is-it-bad-for-people-and-wildlife>

<https://www.nationalgeographic.com/environment/article/deforestation>